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Past Papers Nat 5 Chemistry

2016 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/100)	%	
A	72+	72%	34.1%
B	61+	61%	23.3%
C	51+	51%	18.7%
D	46+	46%	7.4%
No award	<46	<46%	16.5%

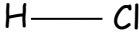
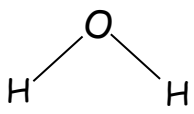
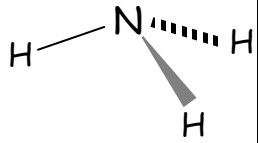
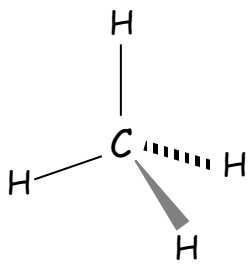
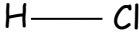
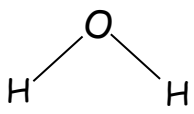
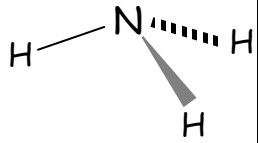
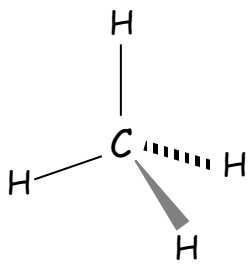
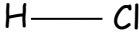
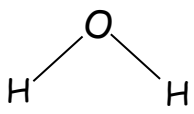
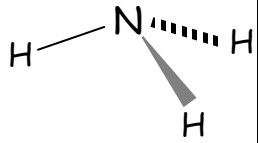
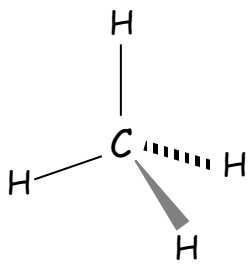
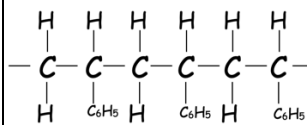
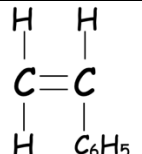
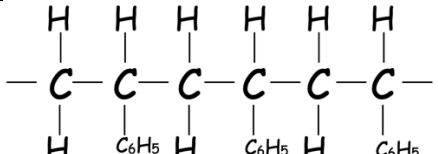
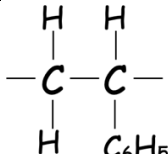
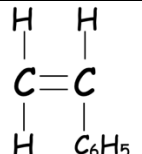
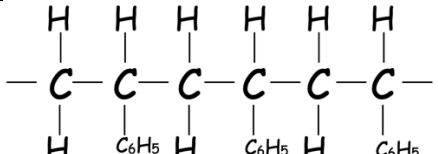
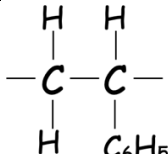
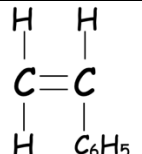
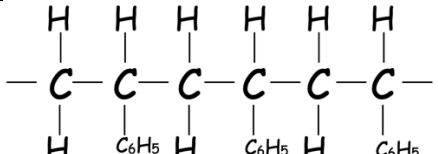
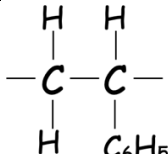
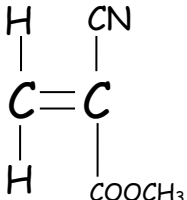
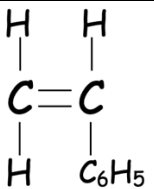
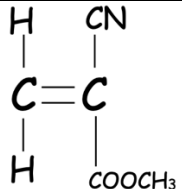
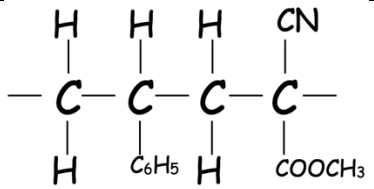
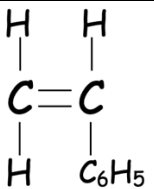
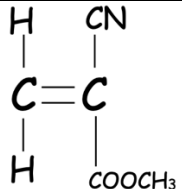
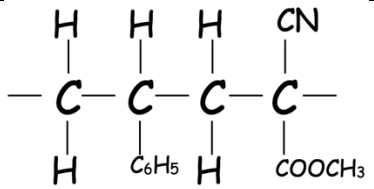
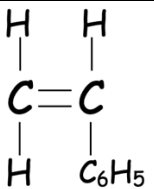
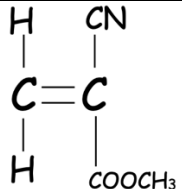
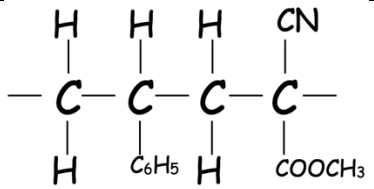
Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	12.4 /20	36.3 /60	14.3 /20

2016 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning
1	D	62	☒ A products of dissolving should be aqueous solutions of $\text{Na}^+(\text{aq})$ and $\text{Cl}^-(\text{aq})$ ions ☒ B correct formula for water should be $\text{H}_2\text{O}(\text{l})$ as water does not dissolve in water ☒ C $\text{NaCl}(\text{aq})$ does not exist as $\text{NaCl}(\text{s})$ splits up its lattice into $\text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$ ions ☒ D $\text{NaCl}(\text{s})$ dissolves in $\text{H}_2\text{O}(\text{l})$ to form the ions $\text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
2	C	79	☒ A temperature would have to be above 40°C to have reaction time less than 10s ☒ B temperature would have to be between $30\text{--}40^\circ\text{C}$ to have time between 10–20s ☒ C temperature is less than 30°C $\therefore$ reaction time must be more than 20s concentration is more than 0.1 $\therefore$ reaction time must be less than 60s ☒ D concentration would have to be 0.1mol l^{-1} and temperature below 20°C
3	D	48	☒ A electrons have negligible mass so removal of electron has no effect on mass no. ☒ B no change to the number of protons so atomic number is unchanged ☒ C no change to the number of protons so charge of nucleus is unchanged ☒ D atom X (e.g. Na 2,8,1) becomes ion X^+ (e.g. Na^+ 2,8) loses an occupied energy level
4	B	77	☒ A sulphur is a non-metal which forms molecules of S_8 with covalent bonds ☒ B copper is a metal which contains metallic bonding ☒ C oxygen is a non-metal which forms molecules of O_2 with covalent bonds ☒ D hydrogen is a non-metal which forms molecules of H_2 with covalent bonds
5	B	78	☒ A ionic compounds contain at least one metal and one non-metal in compound ☒ B monatomic substances are single atoms with no bonds between them ☒ C covalent network compounds e.g. SiO_2 are compounds with many covalent bonds ☒ D covalent molecular compounds e.g. H_2O are compounds with covalent bonds inside
6	C	80	☒ A covalent molecular as no conductivity and low mpt/bpt ☒ B covalent as no conductivity (unclear if network or molecular from mpt) ☒ C ionic bonding as no conductivity as solid but conductivity as a liquid ☒ D metallic bonding as conductivity as both solid and liquid
7	A	45	☒ A silver (I) oxide has the formula Ag_2O ☒ B silver (II) oxide has the formula AgO ☒ C silver (III) oxide has the formula Ag_2O_3 ☒ D silver (IV) oxide has the formula AgO_2
8	C	63	☒ A tin is metal, tin oxide is insoluble $\therefore$ no effect on pH ☒ B zinc is a metal, zinc oxide is insoluble $\therefore$ no effect on pH ☒ C sulphur is a non-metal, sulphur dioxide is a soluble non-metal oxide $\therefore \text{pH} < 7$ (acid) ☒ D sodium is a metal, sodium oxide is a soluble metal oxide $\therefore \text{pH} > 7$ (alkali)
9	B	63	☒ A carbon (soot) is formed by incomplete combustion in a limited supply of air ☒ B carbon dioxide & water formed by complete combustion in plentiful supply of air ☒ C carbon monoxide is formed by incomplete combustion in a limited supply of air ☒ D hydrogen is formed by incomplete combustion in a limited supply of air
10	C	79	☒ A C_4H_8 fits the general formula C_nH_{2n} but C_3H_8 fits the general formula $\text{C}_n\text{H}_{2n+2}$ ☒ B C_4H_8 fits the general formula C_nH_{2n} but C_3H_8 fits the general formula $\text{C}_n\text{H}_{2n+2}$ ☒ C both C_3H_8 and C_5H_{12} fit the general formula $\text{C}_n\text{H}_{2n+2}$ ☒ D C_5H_{10} fits the general formula C_nH_{2n} but C_3H_8 fits the general formula $\text{C}_n\text{H}_{2n+2}$
11	B	75	☒ A side groups cannot be placed on C_1 (side groups must be on middle carbons) ☒ B 5 carbons in main chain with $\text{C}=\text{C}$ bon between C_1 & C_2 and $-\text{CH}_3$ groups on C_2 & C_3 ☒ C $\text{C}=\text{C}$ double bond must be given lowest numbering system ☒ D same numbering system must be used at all times (starting on right here)
12	B	61	☒ A but-1-ene shown is exact same structure as right isomer at top of question ☒ B same formula (C_4H_8) but different structure (methylpropene) ☒ C cyclobutene C_4H_6 has a different formula from butene C_4H_8 ☒ D pent-1,4-diene C_5H_8 has a different formula to butene C_4H_8

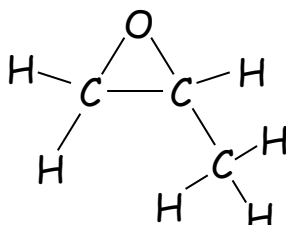
13	A	49	H O H H C C O C H H H 2C carboxylic acid side 1C alcohol side Ester ends in ETHANOATE Ester begins in METHYL																																	
14	C	82	☒A flash point of octane is 15°C ∴ octane does not ignite at 0°C ☒B hexene and cyclohexane have same molecular mass but different flash points ☒C as boiling point increases, flash point increases ☒D as number of carbons increases, the flash point increases																																	
15	D	64	<table><tr><td>Method</td><td colspan="2">Electrolysis</td><td colspan="2">Heat With Carbon</td><td colspan="2">Heat Alone</td></tr><tr><td rowspan="3">Metals Made This Way</td><td>Potassium</td><td>Sodium</td><td>Zinc</td><td>Iron</td><td>Mercury</td><td>Silver</td></tr><tr><td>Lithium</td><td>Calcium</td><td>Nickel</td><td>Tin</td><td>Gold</td><td>Platinum</td></tr><tr><td>Magnesium</td><td>Aluminium</td><td>Lead</td><td>Copper</td><td></td><td></td></tr><tr><td>Reason</td><td colspan="2">most reactive metals</td><td colspan="2">medium reactive metals</td><td colspan="2">least reactive metals</td></tr></table>	Method	Electrolysis		Heat With Carbon		Heat Alone		Metals Made This Way	Potassium	Sodium	Zinc	Iron	Mercury	Silver	Lithium	Calcium	Nickel	Tin	Gold	Platinum	Magnesium	Aluminium	Lead	Copper			Reason	most reactive metals		medium reactive metals		least reactive metals	
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Reason	most reactive metals		medium reactive metals		least reactive metals																															
16	D	20	☒A polyesters are made from two monomers (diacid and diol) ☒B addition monomers contain unsaturated C=C double bonds ☒C monomers with one functional group do not extend the length of the polymer ☒D each monomer in polyester has two functional groups (diacid and diol)																																	
17	A	70	☒A a radioisotope which is alpha emitting and has a long half life ☒B gamma rays are too penetrating to be stopped by a smoke ☒C a smoke detector with a short half life would need to be replace too often ☒D gamma rays are too penetrating to be stopped by a smoke																																	
18	A	50	234 90 Th → 234 91 Pa + 0 -1 e																																	
19	C	45	<table><tr><td>Time (years)</td><td>% ¹⁴C Content</td></tr><tr><td>0</td><td>100</td></tr><tr><td>5600</td><td>50</td></tr><tr><td>11200</td><td>25</td></tr></table>	Time (years)	% ¹⁴ C Content	0	100	5600	50	11200	25																									
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20	B	50	☒A no precipitate as lithium carbonate is soluble ☒B turns cloudy as insoluble calcium carbonate precipitate is formed ☒C no precipitate as sodium carbonate is soluble ☒D no precipitate as ammonium carbonate is soluble																																	

2016 National 5 Chemistry Marking Scheme

Long Qu	Answer	Reasoning																				
1a(i)	<table border="1"><tr><td>1</td><td></td></tr><tr><td></td><td>0</td></tr></table>	1			0	<table border="1"><tr><th>Particle</th><th>Location</th><th>Charge</th><th>Mass</th></tr><tr><td>Proton</td><td>in nucleus</td><td>+1</td><td>1 amu</td></tr><tr><td>Neutron</td><td>in nucleus</td><td>0</td><td>1 amu</td></tr><tr><td>Electron</td><td>outside nucleus</td><td>-1</td><td>approx zero</td></tr></table>	Particle	Location	Charge	Mass	Proton	in nucleus	+1	1 amu	Neutron	in nucleus	0	1 amu	Electron	outside nucleus	-1	approx zero
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1a(ii)	<table border="1"><tr><td>Electron</td><td></td><td>-1</td></tr></table>	Electron		-1																		
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1b	14.5	Relative atomic mass is the average mass of the different isotopes of the same element in a sample. As there are equal quantities of ¹⁴ N and ¹⁵ N in the sample, the RAM is half way between 14 and 15.																				
1c(i)	pyramidal	<table><tr><td> HCl  linear </td><td> H2O  angular </td><td> NH3  trigonal pyramidal </td><td> CH4  tetrahedral </td></tr></table>	HCl  linear	H2O  angular	NH3  trigonal pyramidal	CH4  tetrahedral																
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1c(ii)	Haber Process	Nitrogen + Hydrogen $\xrightarrow{\text{iron catalyst}}$ Ammonia																				
2a		<table><tr><td>  Monomer </td><td>  Polymer </td><td>  Repeating Unit </td></tr></table>	 Monomer	 Polymer	 Repeating Unit																	
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2b		<table><tr><td>  Monomer 1 </td><td>  Monomer 2 </td><td>  Co-polymer </td></tr></table>	 Monomer 1	 Monomer 2	 Co-polymer																	
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3a	Exothermic	<table><tr><td>Exothermic</td><td>Chemical reaction where heat energy is released to surroundings (temperature of the surroundings increases)</td></tr><tr><td>Endothermic</td><td>Chemical reaction where heat energy is absorbed from surroundings (temperature of the surroundings decreases)</td></tr></table>	Exothermic	Chemical reaction where heat energy is released to surroundings (temperature of the surroundings increases)	Endothermic	Chemical reaction where heat energy is absorbed from surroundings (temperature of the surroundings decreases)																
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3b(i)	0.85	$\text{Rate} = \frac{\Delta \text{quantity}}{\Delta \text{time}} = \frac{29 - 12}{30 - 10} = \frac{17}{20} = 0.85 \text{ cm}^3 \text{ s}^{-1}$																				
3b(ii)	60	Maximum volume of hydrogen released = 37cm ³ Time at which 37cm ³ is achieved = 60s																				
3b(iii)	increased reaction rate	Zinc powder has a lower particle size than zinc lumps Lower the particle size the faster the chemical reaction.																				
3c(i)	2Al(NO ₃) ₃	2Al + 6HNO₃ → 2Al(NO₃)₃ + 3H₂ metal + acid → salt + hydrogen																				
3c(ii)	0.36	2Al + 6HNO₃ → 2Al(NO₃)₃ + 3H₂ 2mol 6mol 2mol 3mol 0.01mol 0.015mol 1mol gas = 24litres 0.015mol gas = 24litres × $\frac{0.015}{1}$ = 0.36litres																				

4a(i)	Andalusite + Kyanite	Both andalusite and kyanite can both exist at temperature of 400°C but which mineral is dependent on the pressure.																
4a(ii)	<table><tr><td>Temperature</td><td>500</td></tr><tr><td>Pressure</td><td>4</td></tr></table>	Temperature	500	Pressure	4	The only temperature and pressure where all three forms of the mineral exist is the point on the graph where all three minerals intersect i.e. 500°C and 4kbar pressure.												
Temperature	500																	
Pressure	4																	
4b	17.3	$\text{gfm Al}_2\text{SiO}_5 = (2 \times 27) + (1 \times 28) + (5 \times 16) = 54 + 28 + 80 = 162\text{g}$ $\% \text{ Si} = \frac{\text{total mass of Si}}{\text{gfm Al}_2\text{SiO}_5} \times 100 = \frac{28}{162} \times 100 = 17.3\%$																
5a	Does not react with water, air, alkalis and most acids	Problem Solving: retrieval of information from written passage																
5b	118	Mass number = 197 (from passage) Atomic number = 79 (from data booklet) Number of neutrons = mass number - atomic number = 197 - 79 = 118 neutrons																
5c(i)	$\text{CO} + \text{O}_2 \longrightarrow \text{CO}_2$	carbon monoxide + oxygen $\longrightarrow$ carbon dioxide Formula is derived directly from name (mono = 1) oxygen is one of the seven diatomic elements Formula is derived directly from name (di = 2) $\downarrow$ $\downarrow$ $\downarrow$ CO + O₂ $\longrightarrow$ CO₂																
5c(ii)	catalyst	A catalyst speeds up a chemical reaction but is not used up in the reaction.																
5d	hydrogen hydroxide	Nanorods are grown in a dilute solution of auric acid: <table><tr><th>Solution Type</th><th>Description</th></tr><tr><td>Acidic Solution</td><td>Number of H⁺ ions > Number of OH⁻ ions</td></tr><tr><td>Neutral Solution</td><td>Number of H⁺ ions = Number of OH⁻ ions</td></tr><tr><td>Alkaline Solution</td><td>Number of OH⁻ ions > Number of H⁺ ions</td></tr></table>	Solution Type	Description	Acidic Solution	Number of H ⁺ ions > Number of OH ⁻ ions	Neutral Solution	Number of H ⁺ ions = Number of OH ⁻ ions	Alkaline Solution	Number of OH ⁻ ions > Number of H ⁺ ions								
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6a(i)	Potassium gives a lilac flame test	<table><tr><td>Element</td><td>Barium</td><td>Calcium</td><td>Copper</td><td>Lithium</td><td>Potassium</td><td>Sodium</td><td>Strontium</td></tr><tr><td>Flame Colour</td><td>green</td><td>orange-red</td><td>blue-green</td><td>red</td><td>lilac</td><td>yellow</td><td>red</td></tr></table>	Element	Barium	Calcium	Copper	Lithium	Potassium	Sodium	Strontium	Flame Colour	green	orange-red	blue-green	red	lilac	yellow	red
Element	Barium	Calcium	Copper	Lithium	Potassium	Sodium	Strontium											
Flame Colour	green	orange-red	blue-green	red	lilac	yellow	red											
6a(ii)	Ammonium contains nitrogen	Fertilisers contain soluble compounds of one or more of the following elements <table><tr><td>Nitrogen</td><td>Phosphorus</td><td>Potassium</td></tr></table>	Nitrogen	Phosphorus	Potassium													
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6b	$\text{H}_3\text{PO}_4 + 3\text{NH}_4\text{OH}$ $\downarrow$ $(\text{NH}_4)_3\text{PO}_4 + 3\text{H}_2\text{O}$	$\text{H}_3\text{PO}_4 + 3\text{NH}_4\text{OH} \longrightarrow (\text{NH}_4)_3\text{PO}_4 + 3\text{H}_2\text{O}$																
7	Open Question answer containing:	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.										
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8a		C=C bond to make each carbon up to four bonds Methyl -CH₃ side group C=C bond to make each carbon up to four bonds $\downarrow$ $\uparrow$ $\downarrow$ CH₂C(CH₃)CHCH₂																
8b		The vapours from the steam distillation pass through the delivery tube into the test tube. The cold surface of the test tube allows the vapours to condense. The essential oils will float on top of the water in the test tube.																

8c(i)	addition	Bromine adds across a C=C double bond in an addition reaction H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H H
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11d(ii)		Epoxides have a ring of three atoms but one of the three atoms in the ring must be the oxygen resulting in the third carbon being outside the ring.						
12a	Hydroxyl group	<table><tr><td>—O—H</td><td>$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$</td></tr><tr><td>hydroxyl group</td><td>carboxyl group</td></tr></table>	—O—H	$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$	hydroxyl group	carboxyl group		
—O—H	$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$							
hydroxyl group	carboxyl group							
12b	Weak Acid Strong Base	Betanin is the indicator use $\therefore$ pH range of colour change = 9.0 - 10.0 pH at end point is above 7 weak acid and strong base used in reaction (oxalic acid) (sodium hydroxide)						
12c	15.0cm ³	Average Volume = $\frac{14.9 + 15.1}{2} = \frac{30.0}{2} = 15.0\text{cm}^3$ NB: Ignore rough titre in the calculation of the average volume as it is inaccurate						
12d	0.02	no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{1.8}{90} = 0.02 \text{ mol}$						
13	Open Question answer containing:	3 mark answer	2 mark answer	1 mark answer				
		Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.				

2016 National 5 Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D
1	D	9	19	9	62
2	C	5	13	79	3
3	D	13	10	29	48
4	B	7	77	6	9
5	B	4	78	10	8
6	C	5	5	80	10
7	A	45	51	1	3
8	C	7	5	63	24
9	B	4	63	12	20
10	C	6	7	79	7
11	B	18	75	4	3
12	B	28	61	8	3
13	A	49	37	8	5
14	C	2	3	82	13
15	D	13	11	12	64
16	D	19	44	16	20
17	A	70	4	14	11
18	A	50	10	35	5
19	C	35	9	45	11
20	B	11	50	15	24